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B.Sc. Engg. SWE 1st Semester 15 December 2022

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
Department of Computer Science and Engineering (CSE)

SEMESTER FINAL EXAMINATION **WINTER SEMESTER, 2021-2022**

DURATION: 3 HOURS **FULL MARKS: 150**

CSE 4107: Structured Programming I

Programmable calculators are not allowed. Do not write anything on the question paper.
Answer all **6 (six)** questions. Marks of each question and corresponding CO and PO are written in the right margin with brackets.

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1. a) Explain why using an '`if` statement' is preferable to a '`switch` statement'. 6 (CO2) (PO1)

   b) Assume that `x` and `y` are variables of `float` datatype and `z` is a variable of `int` datatype in a program. In that program the `scanf` function is called as follows: 6 (CO2) (PO1)
   ```
   scanf("%f%d%f",&x,&z,&y);
   ```
   If the user enters the following inputs:
   ```
   12.3  45.6  789
   ```
   Determine the values that will be stored in the `x`, `y`, and `z` variables with proper justification.

   c) The **"Yay Festival"** takes place on February 29 in leap years. In Gregorian calendar, each leap year has 366 days instead of 365, by extending February to 29 days rather than the common 28 days. These extra days occur in each year that is an integer multiple of 4 (except for years divisible by 100, but not by 400). 13 (CO3) (PO2)
   Write a C program to assess whether or not a certain year is appropriate for staging the **"Yay Festival"**.

2. a) Assume that i is an `int` variable and f is a `float` variable. Determine the type of the conditional expression $(i > 0 \ ? \ i \ : \ f)$ with proper justification. 6 (CO2) (PO1)

   b) An insurance company has devised the following rules to calculate the premium and insurance amount for an individual: 13 (CO3) (PO2)
   - If a person's health is excellent and the person is between 25 and 35 years of age and lives in a city and is a male, then the premium is $4 per thousand and his insurance amount will be $2,00,000.
   - If a person satisfies all the above conditions except that the gender is female, then the premium is $3 per thousand and her insurance amount will be $2,50,000.
   - If a person's health is poor and the person is between 25 and 35 years of age and lives in a village and is a male, then the premium is $6 per thousand and his insurance will be $10,000.
   - In all other cases the person is not insured.

   Write a C program to determine the insurance amount and the premium of a person given his demographic information (i.e., age, gender, location etc.).

c) Explain why we should avoid writing statements as **indicated by the comment** in Figure 1: 6 (CO2) (PO1)

```
int i = 0, N = 38;
int a[N], b[N];
while (i<N)
    a[i] = b[i++];    // Avoid this type of statement
```

Figure 1: Code Snippet for Question 2 (b)

3. a) Provide a concise explanation of the concept of `signed` and `unsigned` integer overflow. 3 + 3 (CO2) (PO1)

b) Explain the rules of arithmetic conversions (both implicit and explicit) with suitable examples. 3 + 3 (CO2) (PO1)

c) Show the output of the program in Figure 2 with brief justification: 13 (CO3) (PO2)

```
#include<stdio.h>

int main(){
    int arr[] = {2, 4, 6, 8, 10, 1, 3, 5, 7, 9};
    int *p, *q, a, b, c, d, e, f, g, h;
    p = &arr[0];
    q = &arr[5];
    a = *p++;
    b = *q++;
    c = *++p;
    d = *++q;
    e = *(p + 1);
    f = *(q + 1);
    g = *p + 1;
    h = *q - 1;
    printf("%d %d\n", *p, *q);
    printf("%d %d %d\n%d %d %d\n%d %d", a, b, c, d, e);
    printf("%d %d %d", f, g, h);
    return 0;
}
```

Figure 2: Code Snippet for Question 3. c)

4. a) Consider the arithmetic sequence, $f(N) = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \ldots\ldots \frac{1}{N}$, where the value of N can range from 5 to 100 inclusive. 13 (CO3) (PO2)
Write a C program to find the summation of the arithmetic sequence using **recursion**.

b) Suppose you are given an array of size N, where $0 < N < 1000$, and a key value K, where $0 < K < 10000$. 6 + 6 (CO3) (PO2)
Write a C program to perform the following:
   i. Find out whether the key value K is present in the array or not. If the key is present, then print `"FOUND"`, otherwise print `"NOT FOUND"`.
   ii. If the key is present, find out the index of the key and print it.

You have to write separate functions to complete task (i) and (ii).

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5. a) The upper triangular matrix has all the elements below the main diagonal as zero. For example, $\begin{bmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 0 & 0 & 6 \end{bmatrix}$ can be considered as an upper triangular matrix. 13 (CO3) (PO2)

Suppose you are given a matrix of size $n \times n$, where $0 < n < 10$. Write a C program to find out whether the matrix is an upper triangular matrix or not.

b) Show the output of the program in Figure 3: 12 (CO3) (PO2)

```
#include<stdio.h>

int main(){
    int x = 7;
    int y = 10;
    int z = 5;
    int result = 0;
    result = ++y - 10 || z - 5 && x++;
    result += y++ - 11 || z++ - 5 && x++;
    result += y + 1 > 11 && (z++ >= 6 || x++);
    printf("result: %d\n", result);
    printf("x: %d\n", x);
    printf("y: %d\n", y);
    printf("z: %d\n", z);
    return 0;
}
```

Figure 3: Code Snippet for Question 5 (b)

6. a) The FIFA world cup 2022 has officially started this month. Now, a lot of information needs to be recorded regarding the teams and players for future reference. As a football enthusiast your friend asked you to take on the task to record everything. He has devised a way to arrange the information like the following: 7 + 3 + 3 (CO3) (PO2)

The **TEAM** information consists of:

- Country
- Number of players (main team players + substitutes)
- Number of coaches (head coach + assistant coaches)
- Number of matches played
- Number of matches won
- Number of matches lost
- Position in the tournament (if the team cannot complete group stages, then it's position will be "groups" and so on)

The **PLAYER** information consists of:

- Name
- Position
- Number of matches played
- Number of passes made
- Number of goals scored
- Is he a main team player or a substitute

Based on the aforementioned information:

i. Devise an efficient way to store the information of the teams and the players.
ii. Write a function to find out how many matches have been drawn by a particular team.
iii. Based on the goal's scored by an individual, write a function to get the player who scored the most goals.

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b) The `"string.h"` header file provides us with several built-in library functions. The `strlen()` function returns the length of a string passed to it. The `strcpy()` function copies one string to another.
Create your own `strlen()` and `strcpy()` functions in a C program and demonstrate the working principles of those two functions. 12 (CO3) (PO2)